Tracer — HTB SOC Sherlock Investigation

PsExec Lateral Movement & Forensic Artifact Analysis

Scenario

The SOC Team suspects that an adversary is lurking in their environment and are using PsExec to move laterally. A junior SOC Analyst specifically reported the usage of PsExec on a WorkStation. Your task is to investigate the malicious activity and extract forensic artifacts to understand the attacker's actions.

Task 1-4: PsExec Execution Analysis

Methodology

To analyze PsExec execution activity, we turned to the **Windows Prefetch** mechanism. Prefetch files (`.pf`) are created by Windows to optimize application loading times. They store valuable forensic information, including:

- **Execution count** — how many times the application has run.
- **Last run time** — the most recent execution timestamp.
- **Other run times** — historical execution timestamps.
- **Associated files** — files accessed during execution.

This artifact is ideal for determining both the frequency and timing of PsExec execution, as the tool would have been loaded into memory each time it ran.

Prefetch Analysis

We located the Prefetch file for the PsExec service binary:

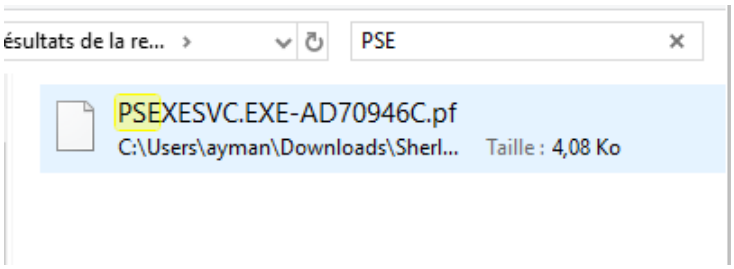


Figure 1: PsExec Prefetch File — PSEXESVC.EXE-AD70946C.pf

We used `PECmd.exe`, a tool for parsing Prefetch files, to extract detailed information:

```
.\PECmd.exe -F "Sherlocks\tracer\Tracer\C\Windows\prefetch\PSEXESVC.
   EXE-AD70946C.pf" --csv "Sherlocks\tracer\Tracer"
```

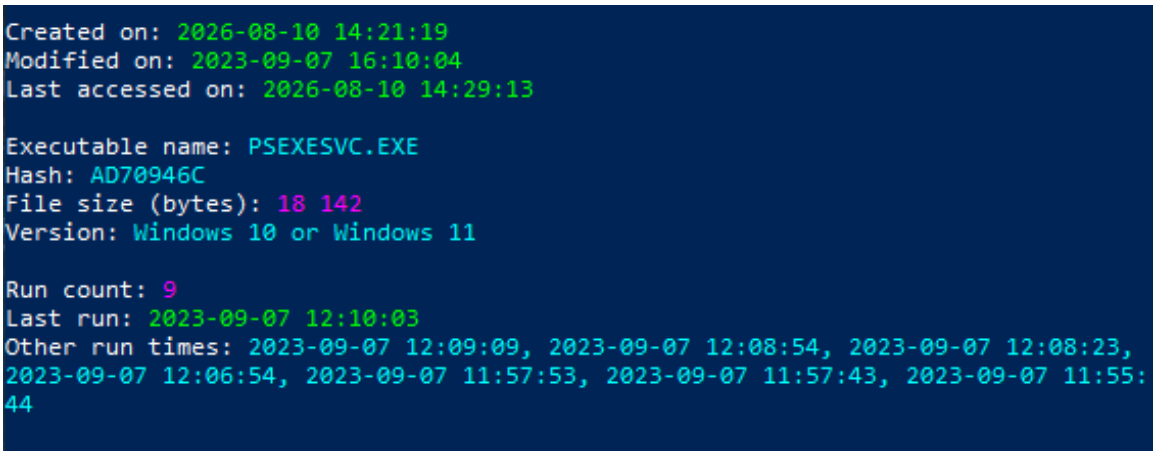


Figure 2: PECmd.exe — PsExec Prefetch Analysis Output

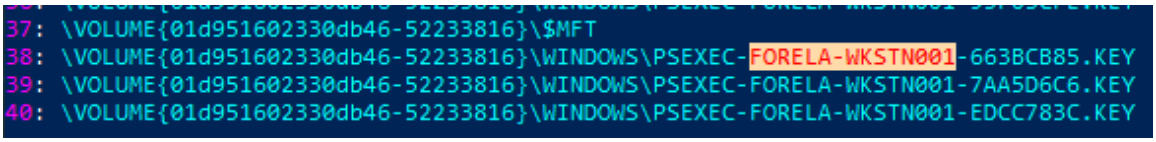


Figure 3: Prefetch File — Referenced Files Showing Hostname

Analysis & Findings

- **Task 1:** How many times was PsExec executed by the attacker on the system?
- **Answer:** `9`

- **Task 2:** What is the name of the service binary dropped by PsExec tool allowing attacker to execute remote commands?
- **Answer:** `PSEXESVC.EXE`

- **Task 3:** We are particularly interested in the 5th last instance of the PsExec. What is the timestamp when the PsExec Service binary ran?
- **Answer:** `2023-09-07 12:06:54`

- **Task 4:** Can you confirm the hostname of the workstation from which attacker moved laterally?
- **Answer:** `FORELA-WKSTN001`

Task 5-6: Key File Analysis

Methodology

To find the key file associated with the 5th last PsExec instance, we needed to examine file system creation timestamps. The **NTFS Journal** (`$J`) records all changes to the file system before they are committed, including:

- File creation, modification, and deletion
- Timestamps and metadata changes
- File name changes

NTFS Journal Analysis

We used `MFTECmd.exe` to parse the `$J` journal file:

```
.\MFTECmd.exe -f 'Sherlocks\tracer\Tracer\C\$Extend\$J' --csv "
   Sherlocks\tracer\Tracer\C\"
```

We then analyzed the output CSV using Timeline Explorer, filtering for PsExec-related key files created around `12:06:54` (the 5th last instance time):

2023-09-07 11:59:23	PSEXEC64.EXE-B77D07A8.pf	DataExtend|DataTruncation
2023-09-07 12:06:55	PSEXEC-FORELA-WKSTN001-95F03CFE.key	FileCreate
2023-09-07 12:06:55	PSEXEC-FORELA-WKSTN001-95F03CFE.key	DataExtend|FileCreate
2023-09-07 12:06:55	PSEXEC-FORELA-WKSTN001-95F03CFE.key	DataExtend|FileCreate|Ren
2023-09-07 12:08:23	PSEXEC-FORELA-WKSTN001-C3E84A44.key	FileCreate

Figure 4: NTFS Journal Analysis — Key File Creation

Analysis & Findings

- Task 5: What is the full name of the Key File dropped by the 5th last instance of PsExec?
Answer: `PSEXEC-FORELA-WKSTN001-95F03CFE.key`

- Task 6: Can you confirm the timestamp when this key file was created on disk?
Answer: `2023-09-07 12:06:55`

Task 7: Named Pipe Analysis

Methodology

PsExec uses **Named Pipes** for interprocess communication between the client and the service. These pipes are created during execution and can provide insight into the specific instance of PsExec.

We analyzed the Sysmon logs for **Event ID 17**, which logs the creation of named pipes:

- **Event ID 17** — Named pipe creation
- Provides pipe name, process path, and timestamp

We searched for Event ID 17 around `2023-09-07 13:06:54` (UTC+1):

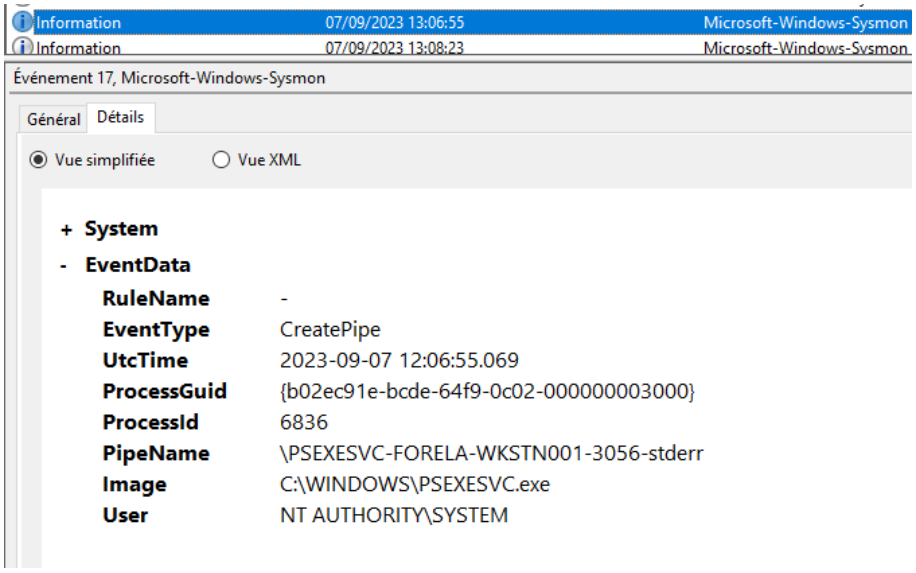


Figure 5: Sysmon Event ID 17 — Named Pipe Creation

Analysis & Findings

- Question: What is the full name of the Named Pipe ending with the "stderr" keyword for the 5th last instance of the PsExec?
- Answer: `\PSEXESVC-FORELA-WKSTN001-3056-stderr`

Attack Chain & Conclusion

The investigation successfully reconstructed the attacker's lateral movement using PsExec:

1. **Lateral Movement Initiated:** Attacker used PsExec from `FORELA-WKSTN001` to move laterally.
2. **Service Binary Dropped:** PsExec dropped `PSEXESVC.EXE` on the target system to establish remote execution.
3. **Execution Frequency:** PsExec was executed 9 times on the system.
4. **Timeline:** The 5th last instance occurred at `2023-09-07 12:06:54`.
5. **Key File Created:** The 5th last instance created `PSEXEC-FORELA-WKSTN001-95F03CFE.key` at `12:06:55`.
6. **Named Pipe Established:** The same instance created the named pipe `\PSEXESVC-FORELA-WKSTN001-3056-stderr` for C2 communication.

The investigation demonstrated the effective use of Windows forensic artifacts to reconstruct the attacker's actions and identify key indicators of compromise.